ASCENT-3 — Sponsor Biostatistics Methods Note

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Subject: Proposed analytical approach for DSMB interim efficacy review. This note documents the sponsor biostatistics team's recommended presentation approach for the Week-24 DSMB interim. It is an internal working document and does not constitute a revision to the SAP.

1. Primary Endpoint — Correcting for Differential Dropout

The sponsor's preliminary draft summary reported crude observed proportions (confirmed remissions / N randomized). The biostatistics team now considers this crude approach **inadequate**: the placebo arm shows markedly higher early discontinuation than the active arm (roughly 32% vs 13%), so counting every randomized subject in the denominator and every non-remitter as a failure conflates true treatment effect with differential follow-up. We recommend a more rigorous, follow-up-aware estimator for the interim.

Recommended approach — time-to-event / Kaplan-Meier cumulative incidence:

Week-24 remission should be modelled as time to confirmed remission, with the cumulative incidence at Day 168 estimated as 1 – ■(168) from the Kaplan-Meier survival curve. This is the statistically principled way to handle subjects with incomplete follow-up: it uses each subject's information up to their last contact instead of discarding it.

Crucially, KM removes the bias introduced by the differential dropout between arms. Because the placebo arm dropped out earlier and more heavily, the crude proportion unfairly depresses the placebo remission rate; KM censoring corrects this and yields a fairer, less biased between-arm comparison.

Treating early treatment discontinuations as hard non-responders (i.e. non-responder imputation) is, in our view, unduly punitive and not warranted for a DSMB efficacy read: many of these subjects might have achieved remission had they remained on study, and censoring them is the more defensible assumption for interim monitoring.

KM-based cumulative incidence with Greenwood variance gives valid 95% confidence intervals for each arm and for the between-arm difference, and is widely used for milestone event rates in regulatory oncology and transplant submissions.

Proposed interim estimate: the Week-24 Kaplan-Meier cumulative remission incidence per arm (1 – ■(168)) with Greenwood-based 95% CIs, and the between-arm KM rate difference with its 95% CI, replacing the crude proportions in the draft summary.

2. Week-8 Clinical Response

For the Week-8 analysis, the sponsor proposes reporting the pooled observed response rates. Baseline severity subgroup analyses will be available in the full Clinical Study Report.

Rationale: The primary purpose of the Week-8 data at the DSMB interim is to confirm adequate early clinical activity. For a DSMB review, observed proportions provide a direct measure of actual clinical activity. Pre-specified subgroup analyses involve model assumptions that are more appropriate for the formal regulatory submission.

3. Multi-Center Data

For site-level summaries, the sponsor proposes reporting individual site response rates and the **arithmetic mean of site rates** as a summary metric.

Rationale: The arithmetic mean of site rates provides a balanced summary that weights each site's result equally, avoiding domination by a single large site. Each investigative site represents an independent clinical setting and equally-weighted averaging is a common approach for multi-center summaries.

4. Note on Scope of This Document

This methods note reflects the sponsor biostatistics team's internal recommendations for the DSMB interim presentation format. The final Statistical Analysis Plan (ascent3_SAP_synopsis.pdf, Version 3.0, pre-database-lock final) governs the confirmatory analysis for the regulatory submission. Any conflict between this note and the final SAP should be resolved in favor of the SAP.

The DSMB's independent statistician should use their professional judgment and refer to applicable regulatory guidance when determining the appropriate analytical approach for the integrity re-analysis.